

# Chapter 4 Nuclear Physics

## Milestones in the Development of Nuclear Physics

- Becquerel discover **radioactivity** in uranium compounds
- Rutherford showed the radiation had three types
- Rutherford scattering experiments
- Nuclear reactions' discovery
$${}^4_2\text{He} + {}^{14}_7\text{N} \rightarrow {}^{17}_8\text{O} + {}^1_1\text{H}$$
- Artificially accelerated protons
- Discover neutron
- Artificial radioactivity
- Nuclear fission
- Controlled nuclear fission

## Some Properties of Nuclei

- All nuclei are composed of **protons and neutrons**(exception ordinary hydrogen)
- **Atomic number**  $Z$  equals the **number of protons** in the nucleus
- **Neutron number**  $N$  equals the number of **neutrons** in the nucleus
- Mass number  $A$  is the **number of nucleons** in the nucleus
$$A = Z + N$$

Symbol

$${}^A_ZX$$

where  $X$  is the chemical symbol of the element.

Same element may contain **varying numbers of neutrons**. They are called the **isotopes**.

## Charge and Mass

Proton:  $+e$ ; Electron:  $-e$ ; Neutron has no charge

Define the mass of one atom of C-12 is  $12u$ , where  $u$  is called the **unified mass unit**.

## Size of Nucleus

Investigated by Rutherford in scattering experiments

Most nuclei are approximately spherical. Average radius is  $r = r_0 A^{1/3}$ .(femtometers, fm)

Density: all nuclei have nearly the same density. The **volume** of the nucleus is proportional to the **total number** of nucleons.

## Nuclear Stability

The nuclei are stable due to the short-range force called the **nuclear force** to attract, which is much stronger than electrostatic force.

If stable:

Light nuclei:  $N = Z$

Heavy nuclei:  $N > Z$

No nuclei:  $Z > 83$

## Binding Energy

The **total energy** of the **bound system** (the nucleus) is **less than** the **combined energy** of the separated nucleons. The **difference** is binding energy.

Binding Energy Per nucleon:

Except for light nuclear, the binding energy per nucleon is stable when  $A > 40$ . Small for light nuclei and large for mid nuclei (peak when  $A = 60$ )

## Radioactivity

The **spontaneous** emission of radiation. The result of **decay or disintegration of unstable nuclei**.

Three types:

- Alpha Particles  
 ${}^4\text{He}$
- Beta Particle  
electron or positrons
- Gamma rays  
high energy photons

| Type               | Composition              | Charge  | Mass       | Penetrating Ability | Ionizing Ability |
|--------------------|--------------------------|---------|------------|---------------------|------------------|
| $\alpha$ particles | ${}^4\text{He}(2p + 2n)$ | $+2e$   | $4u$       | low                 | high             |
| $\beta$ particles  | electrons or positrons   | $\pm e$ | $0.00055u$ | mid                 | mid              |
| $\gamma$ rays      | photons                  | mid     | 0          | high                | low              |

To distinguish them,

- Magnetic field: gamma no deflection, alpha deflected opposite to beta. (If positive is same but the radius of alpha is larger)
- Penetrating experiment
- Ionizing experiment

## The Decay Constant

The number of particles that decay in a given time is **proportional** to the **total number** of particles.

$$\Delta N = -\lambda N \Delta t$$

where  $\lambda$  is called the decay constant.

The decay rate:

$$R = \left| \frac{\Delta N}{\Delta t} \right| = \lambda N$$

Unit: Curie, Ci,  $1\text{Ci} = 3.7 \times 10^{10} \text{decays/second}$ . Bq,  $1\text{Bq} = 1 \text{decays/second}$

The decay curve:

$$N = N_0 e^{-\lambda t}$$

And the half-life is the time takes for half of nuclei to decay

$$T_{1/2} = \frac{\ln 2}{\lambda}$$

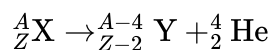
## Decay

The process of one element changing into another element is called **spontaneous decay** or **transmutation**.

The sum of mass number and atomic number are the same on both side of the equation.

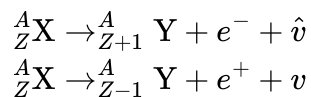
Conservation of mass-energy and momentum hold

## Alpha Decay



The mass of parent nucleus is **larger** than the total mass of He and daughter nucleus. Extra mass are convert into Kinetic energy of particles( $E = \Delta mc^2$ )

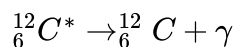
## Beta Decay



$\nu$  is the symbol of neutrino and  $\hat{\nu}$  is antineutrino. (To satisfy the mass conservation)

## Gamma Decay

Excited state to base state



The gamma emission does not change element.

## Uses of Radioactivity

- Carbon Dating
- Smoke detectors
- Radon Pollution

## Natural Radioactivity

Unstable nuclei found in nature gives rise to natural radioactivity (Uranium, Actinium, Thorium).  
Produced in lab through nuclear reactions exhibit artificial radioactivity.

## Nuclear Reactions

Bombarding nuclei with energetic particles causes the change of structure of nuclei. This is called **nuclear reactions**.

Radiation absorbed by matter can cause damage.

## Q Values and Threshold Energy

The energy required to balance a nuclear reaction is **Q value**. (Positive for endothermic reaction, negative for exothermic reaction)

The minimum amount of kinetic energy to conserve the momentum and energy is called the **threshold energy**.

$$KE_{min} = \left(1 + \frac{m}{M}\right) |Q|$$

## Application of Radiation

- Sterilization
- Tracing
- CAT scans

- MRI

## **Radiation Detector**

- Geiger counter: most common form, uses the ionization of a medium
- Semiconductor diode detector: reverse biased p-n junction
- Scintillation counter
- Track detector